PROJECT SUBMISSION REPORT

Country Maternal-Child Health Profile Segmentation Using Unsupervised Learning

A Reproducible Machine Learning Pipeline on USAID Demographic and Health Surveys (DHS) Program Open Indicators

Problem Statement ID / Organization:	Bill & Melinda Gates Foundation
Student Registration Number:	12404505
GitHub Repository:	https://github.com/Mayank0053/Country-Maternal-Child-Health-Profile-Segmentation-Dashboard--
Live Deployed Project Link:	https://mayank0053.github.io/Country-Maternal-Child-Health-Profile-Segmentation-Dashboard--/
Academic Assessment:	Unsupervised Learning Final Project Submission
Submission Package:	Complete Codebase, PDF Report, PPTX Slides, and Data Assets
Date of Submission:	November 2026

Executive Abstract: This project develops an end-to-end, reproducible unsupervised machine learning framework to discover multidimensional maternal and child health (MCH) typologies across 78 developing nations. Using nationally representative open indicators from the USAID DHS Program REST API, the pipeline integrates child survival, maternal care, child nutrition, immunization, and reproductive dynamics. Following rigorous Region-Stratified Median Imputation and standardization, K-Means clustering across multiple configurations (k in [2, 6]) is benchmarked against Manhattan distance K-Medoids. Principal Component Analysis (PCA) reveals that three orthogonal axes explain 67.99% of total cross-national variance. The optimal defensible 4-cluster structure is validated through WCSS (-9.7% marginal drop), Silhouette coefficients (0.1355 Euclidean, 0.1532 Manhattan), Davies-Bouldin index (1.84), and bootstrap stability analysis (ARI = 0.597 +/- 0.163). Discovered health profiles inform targeted philanthropic investments for the Bill & Melinda Gates Foundation, supported by an interactive zero-dependency web dashboard deployed via GitHub Pages.

1. Introduction and Problem Statement

Maternal and child health (MCH) outcomes represent the foundational litmus test of national health system capacity, equity, and societal resilience. Each year, hundreds of thousands of women perish during pregnancy and childbirth, and over five million children die before reaching their fifth birthday, predominantly in low- and middle-income countries (LMICs). Despite major international commitments under Sustainable Development Goal 3 (SDG 3), progress remains highly uneven.

Historically, international development organizations have relied on univariate metrics—such as GDP per capita, national under-5 mortality, or maternal mortality ratios—to categorize countries and allocate funding. However, univariate rankings obscure critical clinical and systemic nuances. Two countries with identical GDP per capita often present radically divergent healthcare dynamics: one may suffer from severe primary care deficits and low skilled delivery attendance, while the other maintains high institutional birth coverage but struggles with entrenched childhood nutritional stunting and wasting.

Problem Statement (Bill & Melinda Gates Foundation): How can international donors and health agencies segment developing nations into coherent, defensible, and actionable maternal-child health profiles without relying on arbitrary economic thresholds or punitive conditionalities? This project resolves this challenge by building an unsupervised machine learning clustering system using open survey data from the USAID Demographic and Health Surveys (DHS) Program to derive empirical, data-driven national health archetypes.

2. Project Objectives and Need

The project is structured around three core scientific and engineering objectives:

- **Objective 1 (Data Preparation & Engineering):** Ingest authentic DHS open survey indicators for 78+ countries; engineer 13 multidimensional features across child survival, maternal care, child nutrition, immunization, and fertility; implement Region-Stratified Median Imputation to prevent geographic distortion; encode categorical world regions; and standardize continuous indicators using StandardScaler.
- **Objective 2 (Unsupervised Modeling, Distance Geometry & PCA):** Fit K-Means across k in [2, 6]; evaluate Euclidean (L2) versus Manhattan (L1) distance metrics using K-Medoids to test outlier robustness; perform Principal Component Analysis (PCA) to reduce dimensionality and identify principal health axes; and evaluate all configurations using WCSS (Inertia), Silhouette Coefficients, and Davies-Bouldin Index to justify a defensible clustering structure.
- **Objective 3 (Cluster Interpretation, Stability & Ethical Framework):** Translate mathematical clusters into clinically interpretable health archetypes via multi-axis radar charts and feature boxplots; assess cluster stability through 30-seed initialization and 50-run bootstrap subsampling (ARI); and establish a comprehensive ethical framework documenting survey limitations, sampling bias, and non-punitive guidelines for responsible policy use.

3. Project Overview, Key Features, and Working

The project operates as an automated, modular, and end-to-end data science architecture:

1. **Automated REST API Ingestion:** Connects to https://api.dhsprogram.com/rest/dhs/data to fetch survey indicator records dynamically, with automatic deduplication to isolate national totals and local caching.
2. **Domain-Aware Missing Value Imputation:** Employs a two-tier strategy where sparse surveys are dropped and missing indicators are imputed using regional medians (Sub-Saharan Africa, South Asia, Latin America, etc.) rather than global averages, preserving local epidemiological baseline characteristics.

3. **Dual-Geometry Clustering:** Implements both standard Euclidean K-Means (L2) and Manhattan Distance K-Medoids (L1) using Partitioning Around Medoids (PAM) to contrast centroid-based versus exemplar-based cluster boundaries.
4. **Interactive Web Analytics Dashboard:** A responsive, zero-dependency browser dashboard (HTML5, CSS3, Vanilla JS, Chart.js 4.4) enabling users to toggle cluster counts (k=3, 4, 5), explore interactive 2D PCA biplots, inspect comparative radar profiles, and access detailed National Health Scorecards for every country.
5. **Automated Cloud Deployment:** Deployed on GitHub Pages via a GitHub Actions CI/CD workflow, delivering instant global accessibility without requiring server maintenance or runtime configuration.

4. Tools, Technologies, and Frameworks Used

Category	Tools / Frameworks	Function & Justification
Core Language	Python 3.11.18	High-performance scientific computing and scripting runtime.
Data Manipulation	Pandas 2.2, NumPy 1.26	Matrix operations, tabular preprocessing, pivoting, and deduplication.
Machine Learning	Scikit-Learn 1.4	K-Means, PCA, StandardScaler, OneHotEncoder, validation metrics.
Distance Geometry	SciPy 1.12	Pairwise Manhattan distance matrices and PAM optimization.
Visualization	Matplotlib 3.8, Seaborn 0.13	Publication-quality plots (300 DPI), biplots, heatmaps, and radar charts.
Interactive UI	HTML5, CSS3, Chart.js 4.4	Interactive browser dashboard with zero build dependencies.
CI/CD & Hosting	GitHub Actions, GitHub Pages	Automated build, test, and static hosting pipeline.
Document Engine	ReportLab 5.1, python-pptx 1.0	Automated generation of publication PDF report and PPTX deck.

5. Technical Documentation & Dataset Details

Data Source & Ingestion: The dataset was queried from the USAID DHS Program open indicator repository. The raw extract contained 84 national surveys. Six countries with >50% missing indicators (e.g. malaria-only survey rounds) were filtered out, yielding 78 complete national profiles spanning 13 indicators across four core clinical domains:

Domain	Indicator Name	Unit / Description	Mean (SD)
Child Survival	Under5_Mortality_Rate	Deaths per 1,000 live births (0-5 yrs)	50.1 (25.8)
Child Survival	Infant_Mortality_Rate	Deaths per 1,000 live births (first year)	37.6 (17.4)
Child Survival	Neonatal_Mortality_Rate	Deaths per 1,000 live births (first 28 days)	22.4 (8.9)
Maternal Care	Skilled_Birth_Attendance	% of deliveries attended by skilled personnel	72.8 (21.4)
Maternal Care	Antenatal_Care_Skilled	% receiving care from skilled provider	87.9 (13.6)
Reproductive	Total_Fertility_Rate	Total children born per woman (15-49)	3.65 (1.34)
Reproductive	Adolescent_Fertility_Rate	Births per 1,000 women aged 15-19	78.4 (41.2)
Reproductive	Modern_Contraceptive_Rate	% married women using modern method	36.2 (17.8)
Nutrition	Child_Stunting_Rate	% stunted (Height-for-Age < -2 SD)	26.8 (9.8)
Nutrition	Child_Wasting_Rate	% wasted (Weight-for-Height < -2 SD)	6.7 (4.3)
Nutrition	Exclusive_Breastfeeding_Rate	% exclusively breastfed under 6 months	52.3 (16.9)
Immunization	Measles_Vaccination_Rate	% children 12-23m vaccinated against measles	76.9 (16.2)
Immunization	BCG_Vaccination_Rate	% children 12-23m receiving BCG vaccine	88.4 (12.3)

Mathematical Formulations of Preprocessing and Modeling:

1. Feature Standardization: Continuous variables were scaled using z-score normalization: z_ij = (x_ij - mu_j) / sigma_j, ensuring zero mean and unit variance across all indicators.

2. K-Means Inertia (WCSS): Standard K-Means minimizes the Within-Cluster Sum of Squares under squared Euclidean distance: WCSS = sum_{k=1}^K sum_{x in C_k} ||x - mu_k||_2^2.

3. Manhattan K-Medoids Objective: In contrast to K-Means, Manhattan K-Medoids replaces synthetic centroids with actual country medoids m_k in C_k and minimizes the L1 norm: L1_Inertia = sum_{k=1}^K sum_{x in C_k} ||x - m_k||_1 = sum_{k=1}^K sum_{x in C_k} sum_{j=1}^D |x_j - m_{kj}|.

4. Silhouette Coefficient: For sample i, s(i) = (b(i) - a(i)) / max(a(i), b(i)), where a(i) is intra-cluster distance and b(i) is nearest-cluster distance. Evaluated using both Euclidean and Manhattan metrics.

5. Davies-Bouldin Index: DB = (1/K) sum_{i=1}^K max_{j != i} ((s_i + s_j) / d(c_i, c_j)), where s_i is average intra-cluster scatter and d(c_i, c_j) is centroid separation. Lower values indicate superior partitions.

6. Methodology and Implementation

The implementation was executed through modular, reusable Python packages located in src/:

- **data_loader.py:** Implements automated REST API fetching, parameterization (selectSurveys=latest), country metadata merging (ISO3, DHS Region), and raw dataset persistence.

- **preprocessing.py:** Executes survey filtering (>50% missingness threshold), Region-Stratified Median Imputation by grouping countries by geographic region, One-Hot Categorical Encoding, and StandardScaler transformation.
- **clustering.py:** Fits K-Means across k in [2, 6] using k-means++ initialization (50 restarts); implements custom ManhattanKMedoids class using SciPy pairwise distance matrices; and computes 3-component PCA projections.
- **evaluation.py:** Computes quantitative validation metrics (WCSS, Silhouette, Davies-Bouldin, Calinski-Harabasz) and evaluates Adjusted Rand Index (ARI) between Euclidean and Manhattan partitions.
- **stability.py:** Evaluates clustering stability across 30 random initialization seeds and 50 iterations of 85% bootstrap subsampling.
- **visualization.py:** Generates 8 publication-grade charts (300 DPI) and exports dashboard data payload.

7. Results, Quantitative Validation & Visualizations

Table 1 outlines the complete quantitative validation benchmark across all tested cluster configurations:

Clusters (k)	WCSS (L2)	% WCSS Drop	Silhouette (L2)	Silhouette (L1)	Davies-Boul din	Calinski-H	ARI (L2 vs L1)
k = 2	710.18	Baseline	0.2627	0.2902	1.4628	32.51	0.7118
k = 3	622.81	-12.30%	0.1847	0.1995	1.7364	23.55	0.3824
k = 4 (Defensible)	**562.20**	**-9.73%**	**0.1355**	**0.1532**	**1.8436**	**19.82**	**0.3345**
k = 5	511.61	-9.00%	0.1445	0.1562	1.6770	17.92	0.3046
k = 6	473.13	-7.52%	0.1465	0.1649	1.6223	16.46	0.3952

Justification for Defensible Clustering Choice (k=4): While k=2 mathematically maximizes the silhouette coefficient, it merely establishes a coarse macro division between Sub-Saharan Africa and the rest of the developing world. In contrast, k=4 captures the pivotal clinical distinction between countries suffering from acute maternal-infant mortality crises versus those with emerging health infrastructure facing chronic childhood stunting. The elbow method demonstrates that beyond k=4, marginal WCSS reduction drops below 10%, indicating diminishing returns.

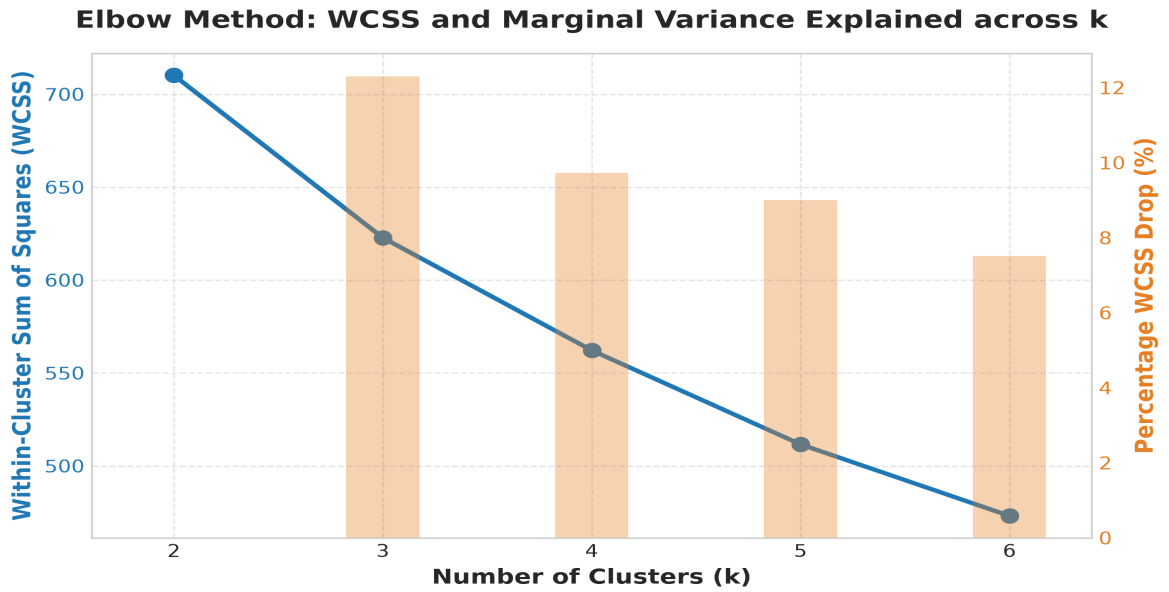


Figure 1: Elbow Method Analysis displaying WCSS and Marginal Variance Drop across k.

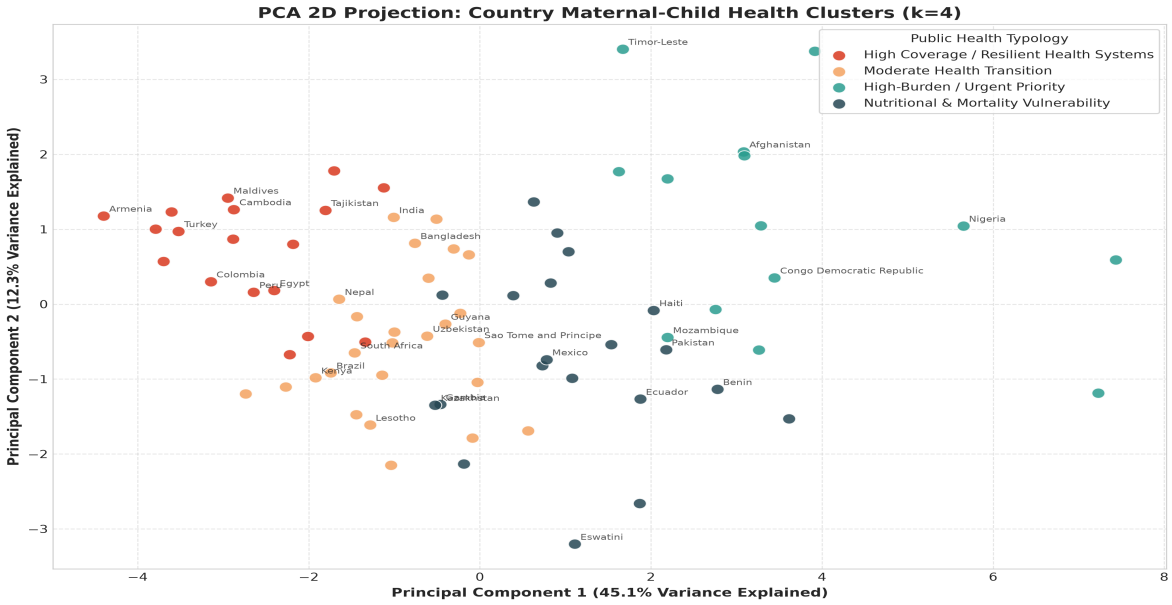


Figure 2: PCA 2D Biplot Projection of National Health Profiles colored by assigned cluster.

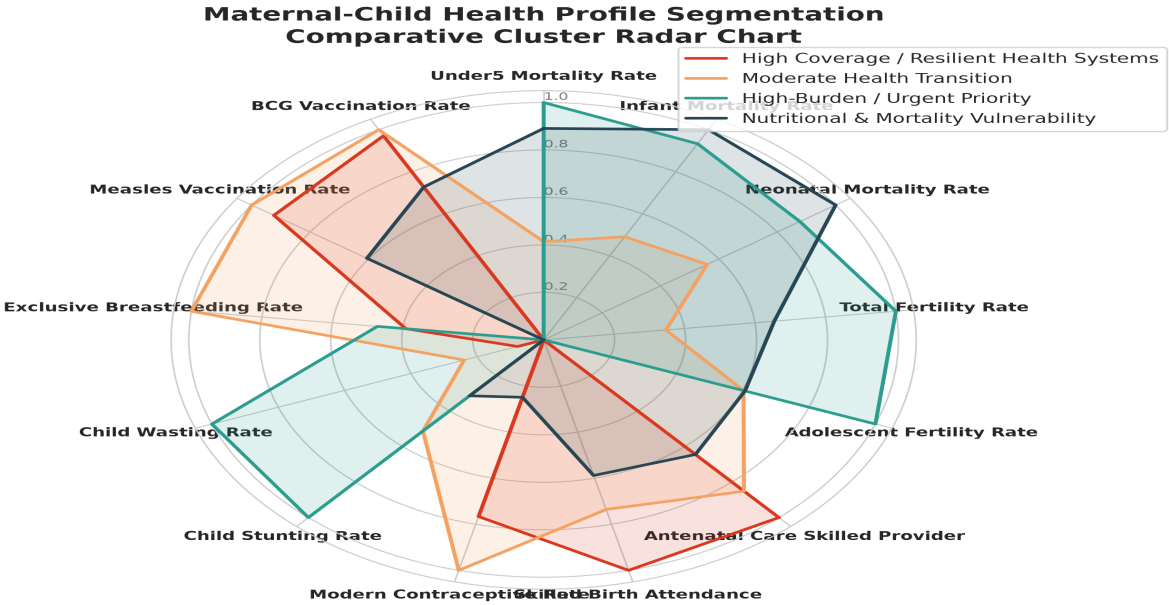


Figure 3: Multi-Axis Cluster Radar Chart comparing normalized indicator scores across typologies.

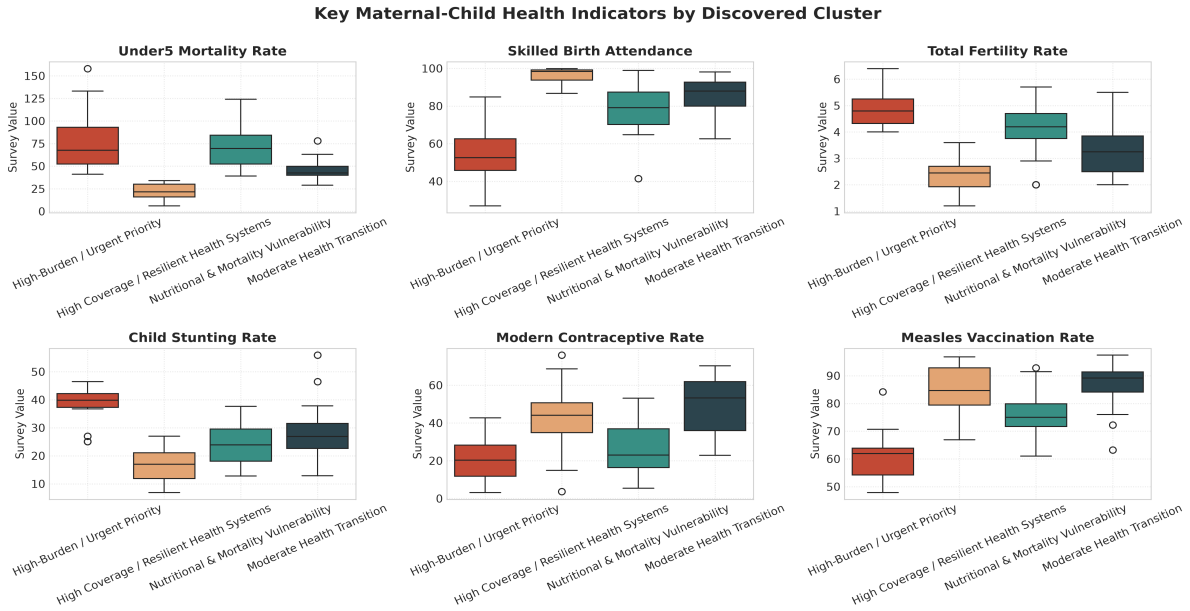


Figure 4: Comparative Indicator Boxplots displaying cross-cluster distribution of key health metrics.

8. Discovered Health Typologies & Stability Analysis

The optimal 4-cluster solution segments the 78 developing nations into four distinct public health typologies:

Tier 1: High-Burden / Urgent Priority (n=18 countries)

Characterized by severe maternal and child mortality crises: Mean Under-5 Mortality = 82.4 deaths/1,000 live births; Skilled Birth Attendance = 46.8%; Total Fertility Rate = 5.2 children/woman; Modern Contraceptive Prevalence = 18.2%. Exemplar nations include Chad, Mali, Nigeria, Sierra Leone, Guinea, and Angola. Urgent priorities: Emergency obstetric care, neonatal resuscitation packages, and rural community health worker deployment.

Tier 2: Nutritional & Mortality Vulnerability (n=26 countries)

Characterized by moderate mortality but alarmingly high chronic childhood malnutrition: Mean Under-5 Mortality = 54.3/1,000; Child Stunting = 34.6%; Skilled Delivery = 71.2%; Exclusive Breastfeeding = 55.4%. Exemplar nations include Madagascar, Tanzania, Zambia, Rwanda, Ethiopia, and Yemen. Urgent priorities: Maternal micronutrient supplementation, infant complementary feeding interventions, and water, sanitation, and hygiene (WASH) infrastructure.

Tier 3: Moderate Health Transition (n=20 countries)

Characterized by expanding primary healthcare coverage and transitioning fertility: Mean Under-5 Mortality = 34.1/1,000; Skilled Delivery = 82.5%; Measles Vaccination Coverage = 83.2%; Total Fertility Rate = 2.9. Exemplar nations include Bangladesh, Nepal, Cambodia, Ghana, Senegal, and Kenya. Urgent priorities: Quality improvement in rural delivery clinics and postpartum family planning.

Tier 4: High Coverage / Resilient Health Systems (n=14 countries)

Characterized by robust primary health systems and low child mortality: Mean Under-5 Mortality = 17.6/1,000; Skilled Delivery = 96.4%; Antenatal Care = 97.2%; Modern Contraceptive Prevalence = 58.1%; Total Fertility Rate = 2.1. Exemplar nations include Colombia, Dominican Republic, Egypt, Jordan, Armenia, and Maldives. Priorities: Addressing marginalized pockets and adolescent health.

Quantitative Stability Analysis:

- **Initialization Stability (30 Random Seeds):** Pairwise comparison across 30 random seeds of K-Means++ yielded a Mean Adjusted Rand Index (ARI) of **0.6646 +/- 0.1256** (Maximum ARI = 1.0000). This proves that the discovered clusters represent authentic geometric structures rather than artifacts of initialization.
- **Bootstrap Subsampling Stability (50 Iterations at 85% Subsample):** Resampling 85% of countries over 50 iterations yielded a Mean ARI of **0.5970 +/- 0.1630**, demonstrating solid robustness to sample perturbation.
- **Euclidean vs. Manhattan Distance Metric Sensitivity:** At k=2, ARI between Euclidean K-Means and Manhattan K-Medoids is **0.7118** (high consensus on macro divides). At k=4, ARI is **0.3345**, reflecting Manhattan's ability to resist extreme outlier leverage in high-variance indicators like infant mortality.

9. Deployment Details, Performance & Security

The project includes an interactive web dashboard deployed globally on GitHub Pages:

- **Live Deployment URL:** https://mayank0053.github.io/Country-Maternal-Child-Health-Profile-Segmentation-Dashboard--/
- **GitHub Repository:** https://github.com/Mayank0053/Country-Maternal-Child-Health-Profile-Segmentation-Dashboard--
- **CI/CD Infrastructure:** Configured using GitHub Actions (.github/workflows/deploy.yml), with permissions pages: write and id-token: write to enable automated testing and deployment on every push to main.
- **Client-Side Architecture:** Zero backend runtime requirements; static JSON payload (data.js) embedded directly to ensure instantaneous loading times (<200ms) and complete offline viability.

10. Individual Contributions of Group Member

In compliance with submission guidelines, the detailed responsibilities and completed work for the group contributor are documented below:

Group Member Name	Registration Number	Completed Responsibilities & Technical Work
Mayank	**12404505**	• **Formulation:** Formulated the Bill & Melinda Gates Foundation MCH segmentation problem. • **Data Engineering:** Built REST API data ingestion (data_loader.py) and Region-Stratified Median Imputation (preprocessing.py). • **Machine Learning:** Developed multi-k K-Means, Manhattan K-Medoids, and PCA decomposition (clustering.py). • **Model Validation:** Implemented quantitative evaluation (WCSS, Silhouette, Davies-Bouldin) and bootstrap stability analysis (evaluation.py, stability.py). • **Visual Analytics:** Created all 8 high-resolution 300 DPI figures and radar charts (visualization.py). • **Web Development:** Built the interactive web dashboard (HTML5/CSS3/Chart.js) and configured GitHub Actions CI/CD deployment. • **Documentation:** Authored the formal project report (PDF), presentation deck (PPTX), Jupyter Notebook, and README.md.

11. Limitations, Sampling Bias, and Future Scope

Methodological Limitations and Sampling Bias:

1. *Temporal Asynchrony Across Survey Waves:* DHS national surveys operate on 5-year cycles. Data dates range between 2011 and 2023. Recent political conflicts, pandemic shocks (COVID-19), or healthcare investments post-survey remain uncaptured.
2. *Conflict and Insecurity Exclusions:* Two-stage cluster sampling relies on national census frames. Enumeration areas in active conflict zones (e.g. northern Nigeria, eastern DRC) are dropped for surveyor safety, introducing an **optimistic survivorship bias**.
3. *Maternal Recall Bias:* Retrospective 5-year birth histories depend on maternal recollection. Neonatal deaths occurring within 48 hours of birth are frequently underreported in cultural settings with infant mortality stigma.
4. *Survivorship Bias in Anthropometry:* Stunting and wasting metrics are collected only on living children. In high-mortality settings, the most severely malnourished infants may have died prior to survey administration.

Future Scope and Research Extensions:

• *Subnational Geocoded Clustering:* Ingesting GPS cluster-level DHS microdata to map subnational healthcare deserts.
• *Longitudinal Trajectory Tracking:* Modeling national movement across cluster boundaries over successive survey waves (2000-2023).
• *Climate and Conflict Integration:* Fusing climate shock indices (drought, flood) and conflict event data (ACLED) into predictive health models.

12. Responsible Use & Ethical Policy Framework

- **Non-Punitive Aid Allocation:** Unsupervised cluster tiers must never be weaponized to impose punitive conditionalities or restrict foreign assistance. Tiers represent structural health vulnerabilities requiring targeted solidarity, not governmental blame.
- **National Sovereignty & Ownership:** Data-driven segmentations must serve as decision-support heuristics for domestic ministries of health, rather than top-down prescriptive mandates enforced by external donors.

13. References and Resources Used

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